

ROC (Receiver Operating characteristic) curve
 → This plot illustrates diagnostic ability of binary classifier

As seen, we have four possible outcomes.

True Positive (TP) → model correctly predicts the positive class

True Negative (TN) → correctly predict negative class

False Positive (FP) → Incorrectly predict +ve class

False Negative (FN) → Incorrectly predict -ve class.

Suppose 1100 samples are tested and found 100 +ve + 1000 -ve.

Our classification model predict 80 as +ve out of 100 + 950 as -ve out of 1000

Predicted class

		Predicted class	
		Positive	Negative
Actual class	Positive	80	20
	Negative	50	950

T.P. → 80

T.N. → 950

F.P. → ~~20~~ 50

F.N. → 20

Sensitivity or Recall or True Positive Rate (TPR)

fraction of correctly identified +ve

$$= \frac{TP}{TP+FN} = \frac{80}{80+20} \sim 0.8$$

Ability to
identify
+ve out of
total +ve

This is vertical axis in ROC.

False Positive Rate (FPR)

~~fraction of incorrectly identified +ve~~

~~= FP~~

fraction of actual negative that
are incorrectly identified as

Positive by the model

$$= \frac{FP}{FP+TN} = \frac{50}{50+450}$$

Misidentification
rate of -ve as
+ve out of total
-ve

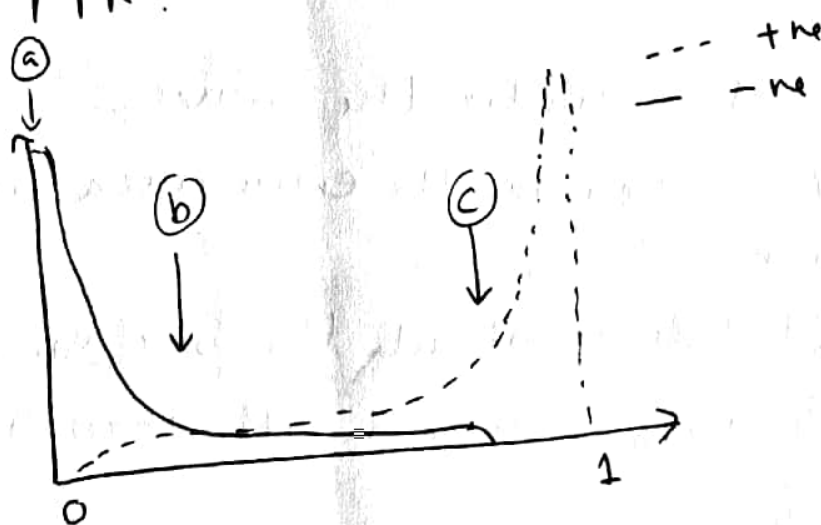
horizontal axis of ROC

$\frac{TP}{TP+FN}$
(TPR)

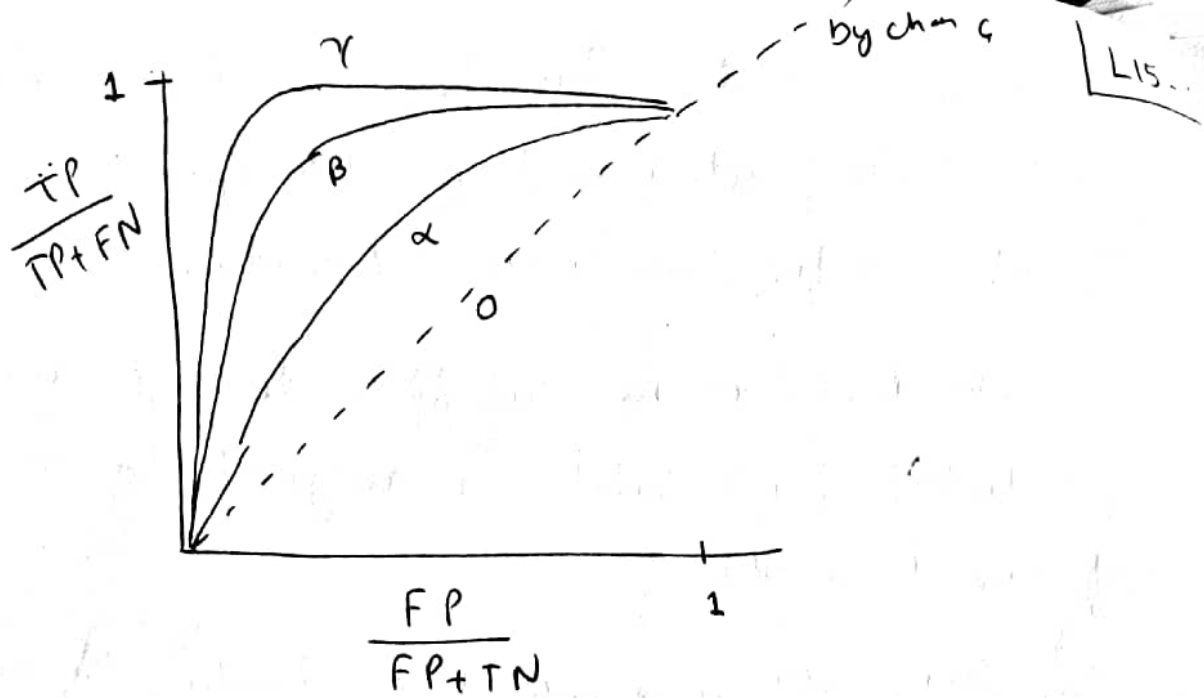
$\frac{FP}{FP+TN}$ (FPR)

ROC is plot of True positive rate (TPR) against False positive rate (FPR) for different classification threshold values.

- Threshold determine cutoff value at which predicted probability is assigned to one of the two classes.
- Lower threshold gives more +ve, leading to higher TPR but also higher FPR.
- Higher threshold gives less +ve, leading to lower TPR & FPR.



- (a) When I put a cut at 0.
all +ve are counted so TPR is 1
also all -ve is +ve so FPR is 1
- (b) Our TPR decrease
However FPR decrease much more.
- (c) Our TPR decrease
but FPR goes to zero!



O is chance 50-50 probability

α is better than O

γ is the best model.

AUC Area under the Curve.

AUC measure the entire area under the ROC curve

It tells us about the performance of the classifier across all thresholds.

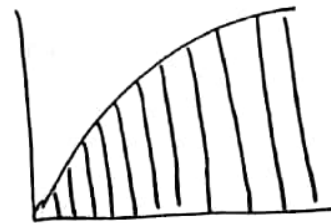
One can also think of AUC as probability that a randomly chosen +ve is ranked higher than randomly chosen -ve.

AUC is integral of ROC curve

$$= \int_0^1 \text{TPR}(\text{FPR}) d(\text{FPR})$$

~~2/2~~

$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$



$$AUC \approx \sum_{i=1}^{n-1} \frac{(x_{i+1} - x_i)(y_i + y_{i+1})}{2} \quad \left[\begin{array}{c} \text{trapezoidal areas} \\ \text{sum} \end{array} \right]$$

$AUC = 1.0 \rightarrow$ Perfect classifier

$AUC > 0.5 \rightarrow$ better than random guessing

$AUC = 0.5 \rightarrow$ Random guessing.

$AUC < 0.5 \rightarrow$ serious issue with the model as it is performing worse than random guessing

AUC + ROC are widely used.

However they are not always the most appropriate especially in cases where dataset is imbalanced.

Imbalanced dataset $\rightarrow$ situation where classes in a classification problem are not represented equally.

One class has more instances than other class.

Imbalance are known to create biases toward majority class, resulting in poor performance especially for the minority class.

$\rightarrow$ Misleading Accuracy

would reach high accuracy for one class (majority) but might fail to even detect minority.

$\rightarrow$ Bias toward majority

$\rightarrow$ Poor Generalization

May overfit majority class and fail to generalize to unseen data and real life problem.

One can use alternative methods and metrics

$\rightarrow$ Precision - Recall (PR) curve

PR curve is useful when one is dealing with imbalanced datasets

Let say +ve is our minority class

Precision is proportion of true positive (TP) among all predicted positive (TP + FP)

$$\frac{TP}{TP + FP}$$

Recall / Sensitivity / True Positive Rate (TPR)

$$= \frac{TP}{TP + FN}$$

PR curve plots Precision against Recall for different thresholds..

Key difference b/w PR curve & ROC curve is that PR curve focuses on positive class and doesn't consider the -ve class

ROC does take or consider -ve.

Useful when no. of -ve instances are more than no. of +ve instances.

While we are more concerned of how well our model will detect +ve.

Similar to AUC; one has Area Under the PR curve (called AUC-PR).

→ PR curves focuses more on how well model behaves with minority class (here +ve)

② F1-score

harmonic mean of precision + Recall

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Useful when one needs a balance b/w precision and recall, especially for imbalanced datasets

F1 score ranges from 0 to 1.

1 → perfect precision + recall

→ F1 score is preferred when false positive + false negatives have different costs.

⊗

→ Doesn't provide insight into model's performance across different thresholds.

③ Accuracy -

Ratio of correctly predicted (both true + -ve) to total no.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

→ useful when classes are equal.

→ Don't use for imbalanced dataset.

Logarithmic loss (Log loss)

Logarithmic loss is used for accuracy of the probabilistic predictions provided by the classifier

It is said to penalize wrong or mis classification heavily and rewards the models that are more confident in their correct prediction

$$\text{log loss} = -\frac{1}{N} \sum_{i=1}^N (y_i \log(p_i) + (1 - y_i) \log(1 - p_i))$$

y_i → actual label (1 for +ve, 0 for -ve)

p_i → predicted probability of the class for i^{th} instance

N → total no. of instances

Useful when one is interested in probability estimates produced by model rather than just the classification.

Matthews Correlation Coefficient (MCC)

MCC is more balanced metric than accuracy, especially for the imbalanced datasets.

$$MCC = \frac{(TP \times TN) - (FP \times FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

MCC values ranges from -1 to 1

-1 → complete disagreement b/w predicted & Actual

0 → Similar to random guessing.

1 → Perfect classification.

Useful in imbalanced classification.

Cohen's Kappa

Cohen's Kappa measures agreement b/w predicted labels and true labels, correcting for the agreement that is expected or could occur by chance.

$$K = \frac{p_o - p_e}{1 - p_e}$$

p_o → observed agreement b/w predicted & true labels
(similar to accuracy)

ok $p_e \rightarrow$ expected agreement by chance

range from 0 to 1.

1: Perfect agreement.

0: Agreement = purely random.

Useful when one wants to account for possibility of agreement by chance, particularly in multi-class classification problems or imbalanced datasets

Let's see an example to understand it more

Two physicians A + B diagnose three diseases α, β, γ .

	B: α	B: β	B: γ	total A
A: α	4	1	0	5
A: β	0	3	1	4
A: γ	1	0	0	1
Total B	5	4	1	10

Observed Agreement p_o is diagonal.

$$p_o = \frac{\text{No. of agreement}}{\text{Total No. of patient}} = \frac{4(\alpha:\alpha) + 3(\beta:\beta) + 0(\gamma:\gamma)}{10} = \frac{7}{10} = 0.7$$

Physician A & B agrees 70% of the diagnoses

Expected Agreement (P_e)

Probability that two physicians would agree on a diagnosis purely by chance

Calculated by taking product of marginal totals (row + column sums)

$$P_e = \frac{(5/10 \times 5/10) + (4/10 \times 4/10) + (1/10 \times 1/10)}{1} = \frac{0.25 + 0.16 + 0.01}{1} = 0.42$$

By chance, one may expect 42% agreement

Cohen's Kappa is calculated as

$$K = \frac{P_o - P_e}{1 - P_e} = \frac{0.7 - 0.42}{1 - 0.42} = \frac{0.28}{0.58} = 0.48$$

Moderate agreement

$K < 0.0$ poor agreement

$0.01 \leq K \leq 0.20 \rightarrow$ slight agreement

$0.21 \leq K \leq 0.40 \rightarrow$ fair agreement

$0.41 \leq K \leq 0.60 \rightarrow$ Moderate agreement

$0.61 \leq K \leq 0.80 \rightarrow$ Substantial agreement

$0.81 \leq K \leq 1.00 \rightarrow$ Almost agreement

im In case the model or person give result in percentage. One can also provide K .

Divide the percentage into categories

Category A 0% to 25%
B 25% to 50%
C 50% to 75%
D 75% to 100%

and do similar

What if we have multiple raters (more than two)

Fleiss' Kappa is used.

It is a statistical measure used to assess the reliability of agreement b/w multiple raters (more than two). when they are classifying items into categories

Further, Fleiss' Kappa can handle situation where not all raters rate all items

→ N items

→ K categories

→ M raters or models

M rater/models assign each item from N items into one of the K categories

~~Items~~

How to calculate Fleiss' Kappa

- ① Calculate the proportion of raters who assigned each category (for all items)

For each category k , calculate proportion of raters who assigned that category across all items.

$$p_k = \frac{\sum_{i=1}^N n_{ik}}{N \times M}$$

p_k → proportion of all ratings that fall into category k .

N → total no. of items

M → no. of raters

n_{ik} → no. of raters that rated item i in category k .

- ② Compute agreement for each item

for each item i , calculate proportion of agreement b/w the raters.

$$p_i = \frac{1}{M(M-1)} \sum_{k=1}^K n_{ik}(n_{ik}-1)$$

M → no. of raters

K → no. of categories

n_{ik} → no. of raters that assigned item i to category k .

p_i → observed agreement for item i

③ Calculate the Overall mean Agreement (Observed Agreement)

$$P_o = \frac{1}{N} \sum_{i=1}^N P_i$$

④ Expected agreement as expected by chance.

$P_e \rightarrow$ probability that raters would agree by chance.

$$P_e = \sum_{k=1}^K p_k^2$$

⑤ Calculate Fleiss' Kappa

$$K = \frac{P_o - P_e}{1 - P_e}$$

$K < 0.0 \rightarrow$ Poor agreement (worse than chance)

$0.01 \leq K \leq 0.20 \rightarrow$ slight agreement

$0.21 \leq K \leq 0.40 \rightarrow$ Fair agreement

$0.41 \leq K \leq 0.60 \rightarrow$ Moderate agreement

$0.61 \leq K \leq 0.80 \rightarrow$ Substantial agreement

$0.81 \leq K \leq 1.00 \rightarrow$ Almost perfect agreement

Example

L15

	A	B	C	D	E
1	1	2	0	1	0
2	3	0	1	0	0
3	2	2	0	0	0
4	0	3	0	1	0
5	1	0	2	1	0
6	0	0	4	0	0
7	2	1	1	0	0
8	1	1	1	1	0
9	0	1	0	2	1
10	0	2	1	0	1

$$① \quad p_k = \frac{\sum_{i=1}^{10} n_{ik}}{N \times M}$$

↓ sum

$$p_A = \frac{1+3+2+0+1+0+2+1+0+0}{10 \times 4} = 0.25$$

$$p_B = \frac{2+0+2+3+0+0+1+1+1+2}{10 \times 4} = 0.30$$

$$p_C = \frac{0+1+0+0+\dots+1}{10 \times 4} = 0.25$$

$$p_D = \frac{6}{10 \times 4} = 0.15$$

$$p_E = \frac{2}{10 \times 4} = 0.05$$

$$192 \quad \textcircled{2} \quad P_i = \frac{1}{M(M-1)} \sum_{k=1}^{K=5} n_{ik}(n_{ik}-1) \quad \boxed{n(n-1)}$$

$$P_1 = \frac{1}{4(4-1)} [1(1-1) + 2(2-1) + 0(0-1) + 1(1-1) + 0(0-1)]$$

$$= \frac{1}{12} (2) = 0.1667$$

$$P_2 = 0.5$$

$$P_3 = 0.3333$$

$$P_4 = 0.5$$

$$P_5 = 0.1667$$

$$P_6 = \frac{1}{4(4-1)} [0(0-1) + 0(0-1) + 4(4-1) + 0(0-1) + 0(0-1)]$$

$$= 1$$

$$P_7 = 0.1667 \quad P_8 = 0.0833, \quad P_9 = 0.1667$$

$$P_{10} = 0.0833$$

$$\textcircled{3} \quad P_0 = \frac{1}{N} \sum_{i=1}^N P_i = \frac{P_1 + P_2 + P_3 + P_4 + P_5 + \dots + P_{10}}{10}$$

$$= 0.3167$$

$$\textcircled{4} \quad P_e = \sum_{k=1}^{K=5} p_k^2 = \cancel{P_A^2 + P_B^2 + P_C^2 + P_D^2 + P_E^2}$$

$$= p_A^2 + p_B^2 + p_C^2 + p_D^2 + p_E^2$$

$$= 0.25^2 + 0.3^2 + 0.25^2 + 0.15^2 + 0.05^2$$

$$= 0.24$$

$$\textcircled{5} \quad K = \frac{P_0 - P_e}{1 - P_e} = \frac{0.3167 - 0.24}{1 - 0.24} = 0.101$$

slight agreement

G - Mean

Geometric Mean is used to evaluate imbalanced datasets; focusing on balancing the performance across classes.

$$G - \text{Mean} = \sqrt{\text{Sensitivity} \times \text{Specificity}}$$

Sensitivity (Recall) $\rightarrow$ proportion of true +ve correctly classified

Specificity $\rightarrow$ proportion of true -ve correctly classified

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{80}{80 + 20} \sim 0.8$$

$$\text{Specificity} = \frac{TN}{TN + FP} = \frac{950}{950 + 50} \sim 0.95$$

Used when one wants to ensure a balance b/w classifying true +ve & -ve correctly, especially in imbalanced datasets.

It focuses on achieving a balance b/w sensitivity and specificity

Qs Youden's Index (J)

Youden's Index (J) is a single statistic which summarizes the performance of binary classifiers

$$J = \text{Sensitivity} + \text{Specificity} - 1$$

→ useful when one want a simple measure of overall performance.

→ It is specific point on the ROC curve that maximize the difference b/w TPR & FPR.

~~It is the point on the ROC curve that~~

while doing multi-class classification.

Computation of ROC + AUC;

two common approaches are used.

1) OvR $\rightarrow$ One-vs-Rest.

This is the most common method.

In this approach, we compute ROC curve for each class separately.

Each class $\rightarrow$ the tested class is treated as +ve while the other are treated as -ve class.

This way we get ROC curve + AUC for each class individually.

Either show them separately or compute an overall average.

2) Macro-Averaging + Micro-Averaging

Once ROC for all classes are gotten.

2 common methods to aggregate results

$\rightarrow$ Macro-Averaging $\rightarrow$ Calculate ROC + AUC for each class & then average them to get a single ROC/AUC.

$\rightarrow$ Micro-Averaging $\rightarrow$ Treat all true +ve, false +ve + false -ve across all classes as a single binary classifier & compute ROC + AUC.

Macro Averaging

First compute ROC curve for each class individually
for each class k
we calculate

True Positive Rate (TPR) Sensitivity.

False Positive Rate (FPR) 1-specificity

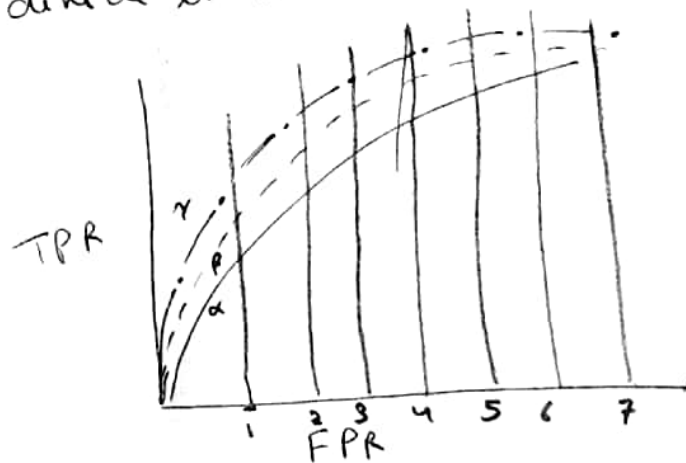
TPR_k , FPR_k

As FPR_k is 0 to 1, divide it into let say 50 equally
space values

Use linear interpolation
and get TPR value for
each FPR value

E.g for 1 get

TPR values for all 3 classes.



~~(TPR_{α}^1 , TPR_{β}^1 , TPR_{γ}^1)~~
~~(TPR_{α}^2 , TPR_{β}^2 , TPR_{γ}^2)~~
~~(TPR_{α}^3 , TPR_{β}^3 , TPR_{γ}^3)~~

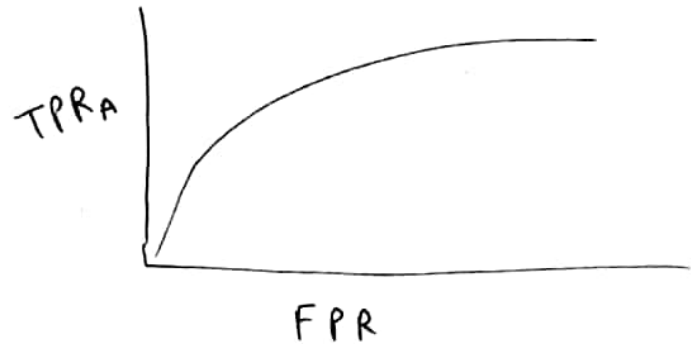
Now

TPR_{α}^1	TPR_{β}^1	TPR_{γ}^1
TPR_{α}^2	TPR_{β}^2	TPR_{γ}^2
TPR_{α}^3	TPR_{β}^3	TPR_{γ}^3
$\vdots$	$\vdots$	$\vdots$
TPR_{α}^7	TPR_{β}^7	TPR_{γ}^7

Now perform average at a given FPR.

$$TPR'_p = \frac{TPR'_\alpha + TPR'_\beta + TPR'_\gamma}{3}$$

$$TPR'_A = \frac{TPR'_\alpha + TPR'_\beta + TPR'_\gamma}{3}$$



Macro Averaged AUC

$$= \frac{1}{K} \sum_{k=1}^K AUC_k$$

AUC_k is area under the ROC curve for class k .

Micro Averaging of the ROC curve

Combines the contributions of all classes to calculate a single ROC.

In micro-averaging approach; one treat every true positive, false positive, and false negative from all classes as if they come from one binary classification problem

① We have 5 classes C_1, C_2, C_3, C_4, C_5

True Positive, F.P.; F.N. of each class estimated

We combine counts of T.P., F.P., F.N. from all classes

$$\text{Total TP} = \sum_{k=1}^K TP_k$$

$$\text{Total TN} = \sum_{k=1}^K TN_k$$

$$\text{Total FP} = \sum_{k=1}^K FP_k$$

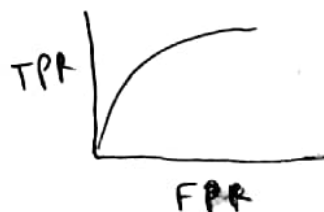
$$\text{Total FN} = \sum_{k=1}^K FN_k$$

$$\text{② True Positive Rate (TPR)} = \frac{\text{Total TP}}{\text{Total TP} + \text{Total FN}}$$

$$\text{False Positive Rate (FPR)} = \frac{\text{Total FP}}{\text{Total FP} + \text{Total TN}}$$

③ Vary Decision threshold and estimate TPR & FPR

AUC estimated by integrating.



How well model can separate the classes based on different thresholds.